FARMGATE AI — GRAIN ASSAY CERTIFICATE

Low-Cost Smartphone Vision Assaying System | AGMARK Standards Compliance Assessment

Sample ID: SMP-0012	**Date/Time:** 2026-10-22 15:31:07
Grain Type / Variety: Basmati	**Capture Quality Status:** Passed

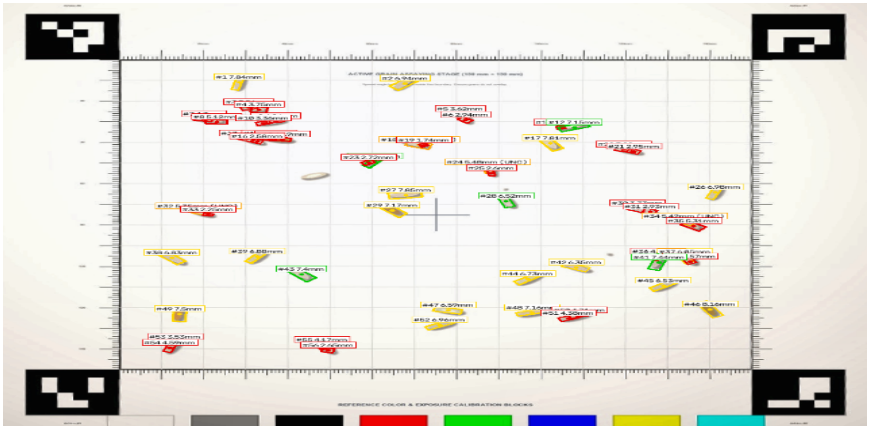
ASSESSED QUALITY GRADE

REJECTION / SUB-STANDARD

Exceeds tolerance limits: Broken grains (53.6%) exceed 25.0% limit; Discoloured grains (30.4%) exceed 6.0% limit.

1. Morphometric & Optical Defect Breakdown

2. Representative Assayed Grain Imagery



3. Assaying Standard Compliance Methodology

Measurements conducted using Farmgate AI high-precision computer vision pipeline. Fiducial perspective rectification performed via OpenCV ArUco (DICT_4X4_50) Homography at sub-millimeter scale. Grain segmentation utilized watershed separation with Otsu-adaptive binarization. Grain dimensional and defect grading adheres to the Directorate of Marketing and Inspection (DMI), Ministry of Agriculture & Farmers Welfare, Government of India (AGMARK Milled Rice Grading and Marking Rules).

Automated Assaying Signature: FARMGATE-AI-VERIFIED-V1.0	**Quality Assurance Officer:** _____ (Sign/Seal)

MANDATORY LEGAL DISCLAIMER: This tool provides image-based preliminary quality estimation. It does not replace certified laboratory testing or measure moisture content, chemical composition, or cooking quality.